Basic Science Department

BAS 141
Engineering Chemistry

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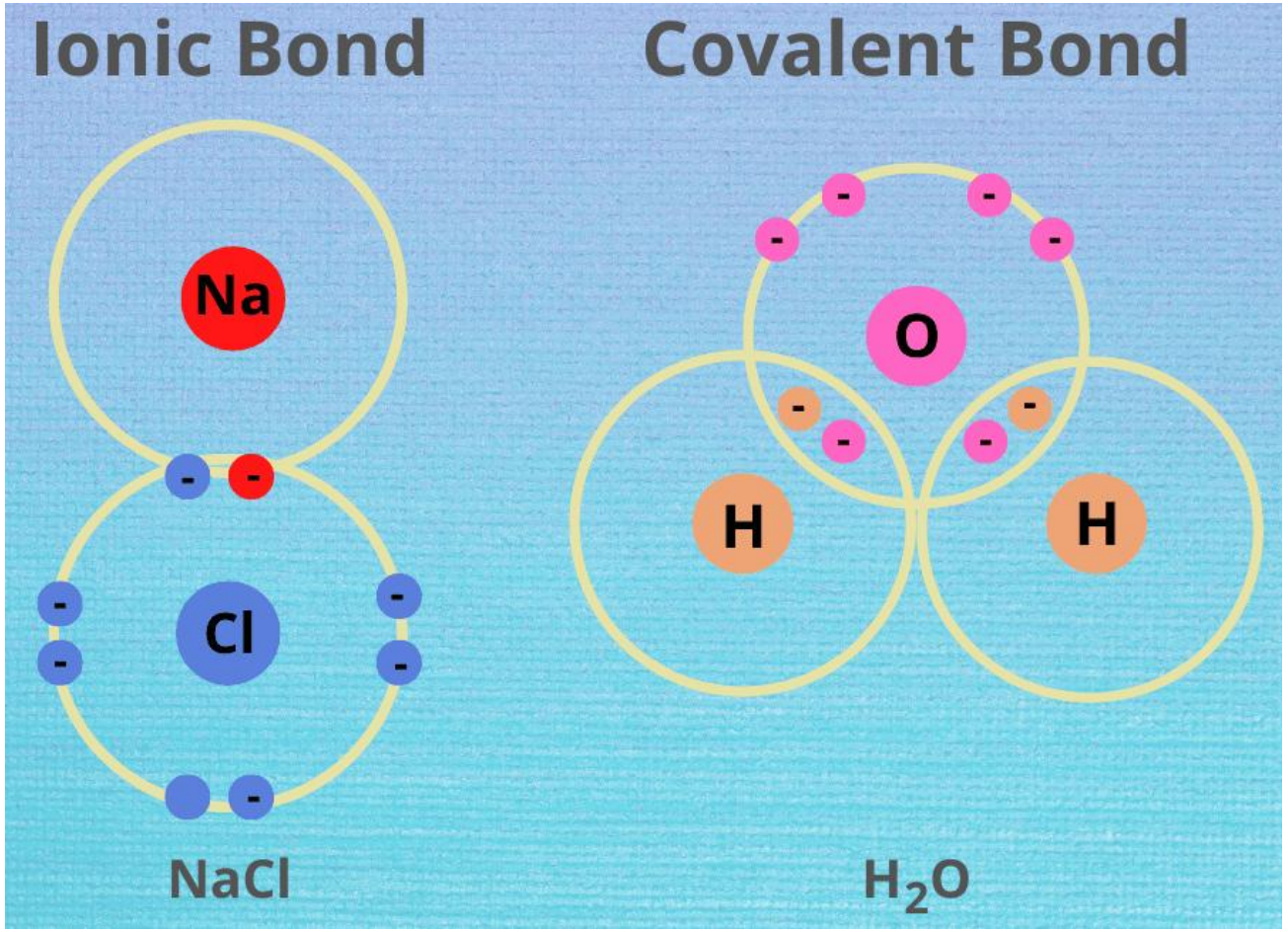
2.1. Intramolecular Forces

Intramolecular forces are Attractive forces that hold atoms together in a molecule

➢**Covalent bond**

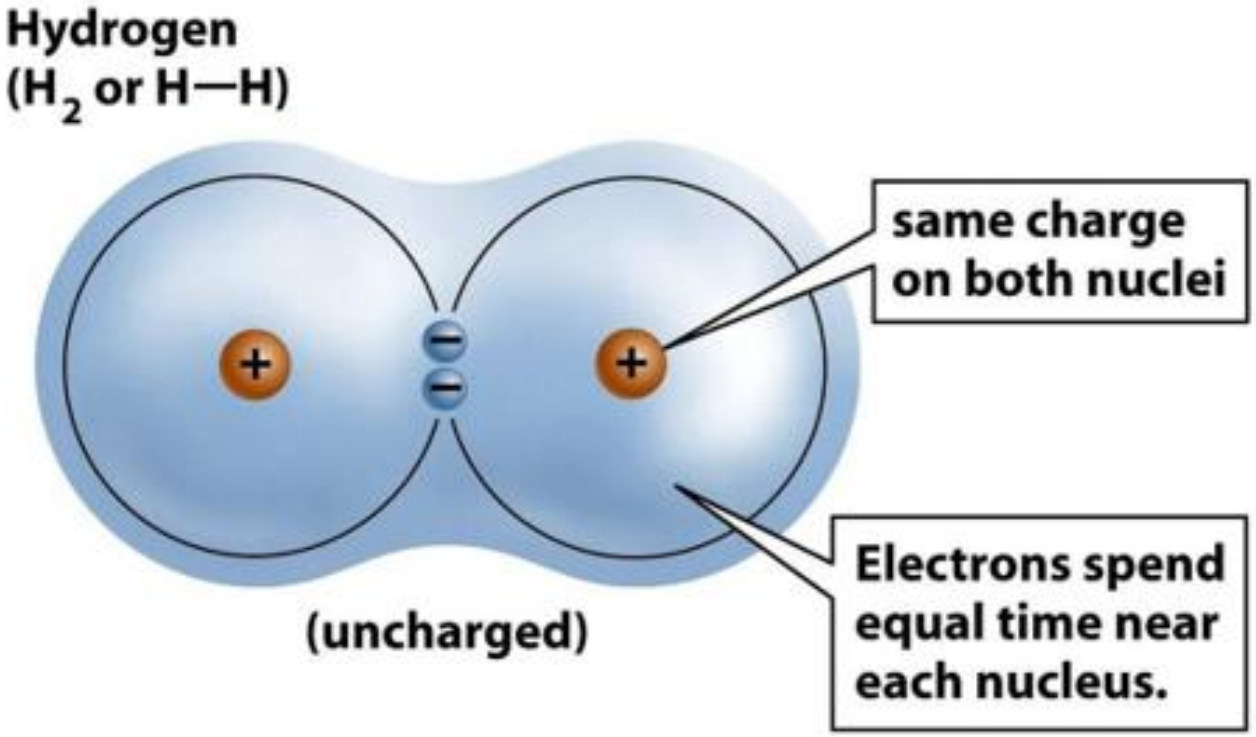
- Nonpolar covalent bond
- Polar covalent bond

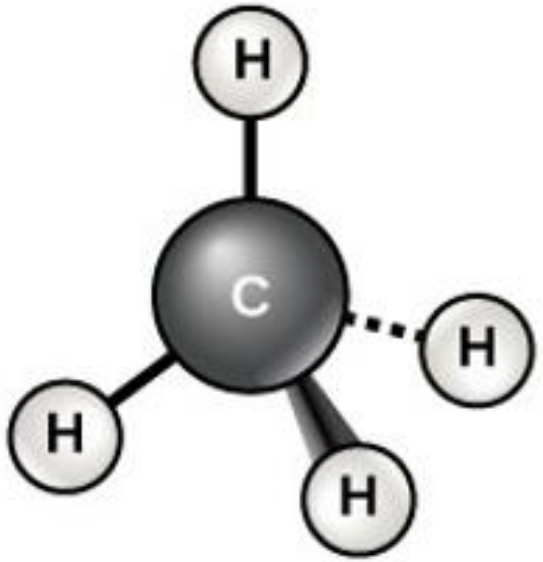
➢**Ionic bond**



➢ Non-polar covalent bond

A bond occurs when two atoms share several (usually two) electrons with a symmetrical distribution of electrons

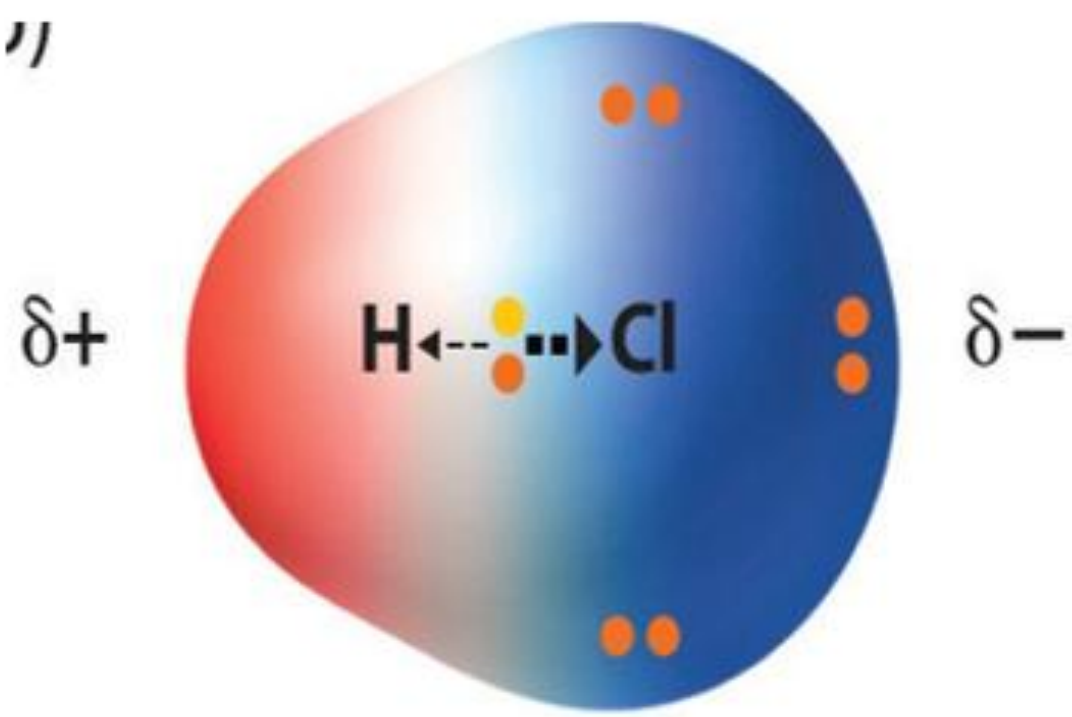


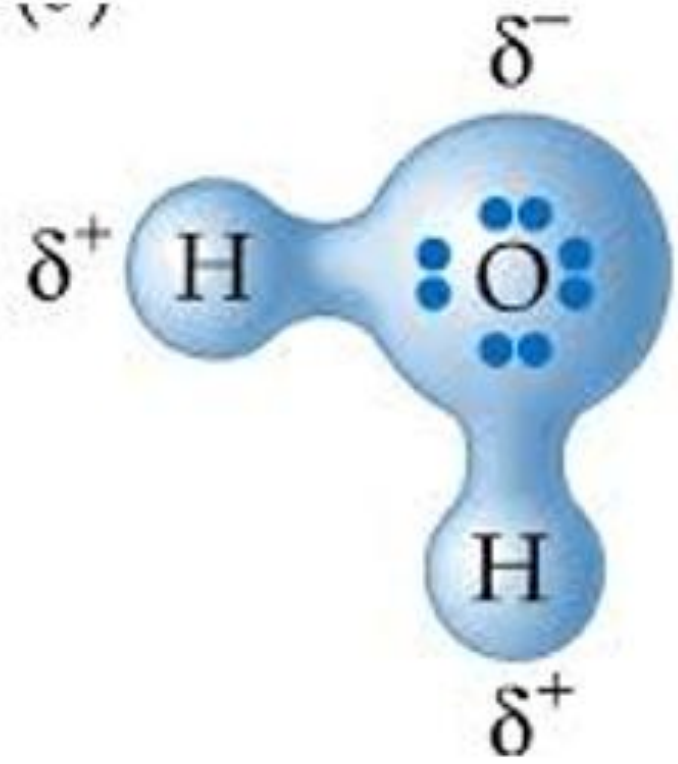


Methane

- **Polar covalent bond**

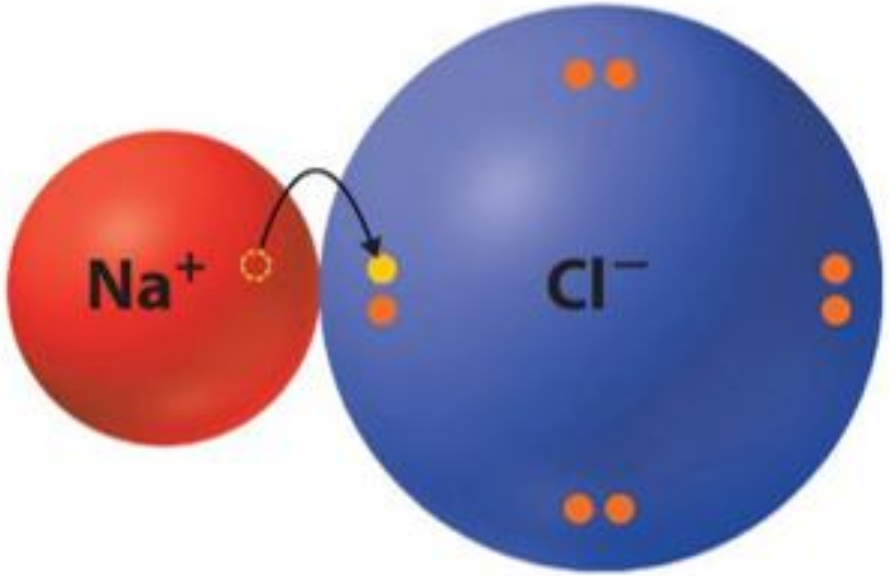
Form between atoms of different electronegativity

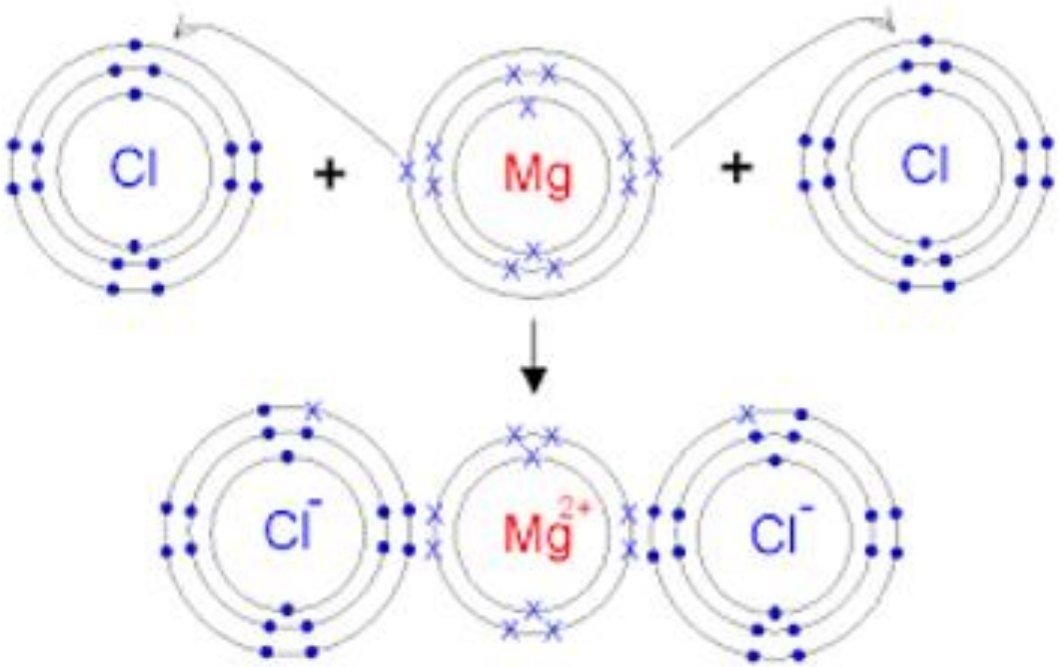




➢ Ionic bond

A bond that results from a complete transfer of one or more electrons between atoms





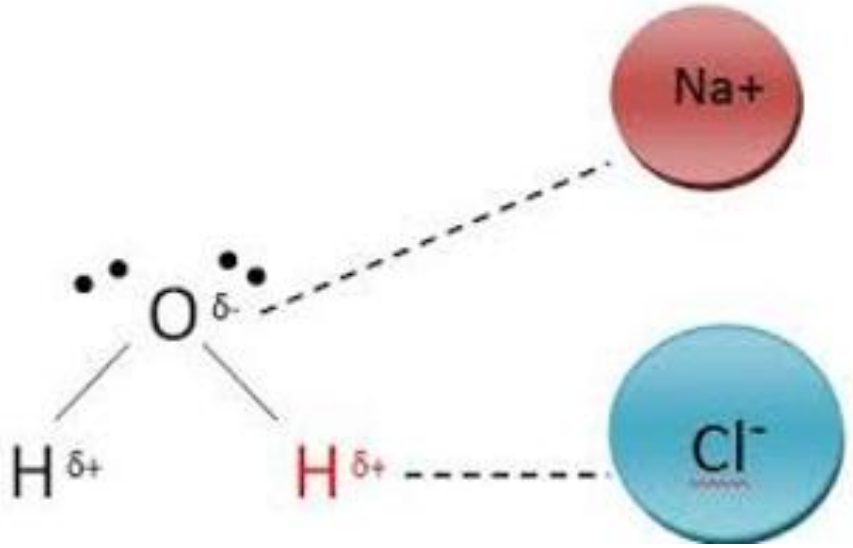
2.2. Intermolecular Forces

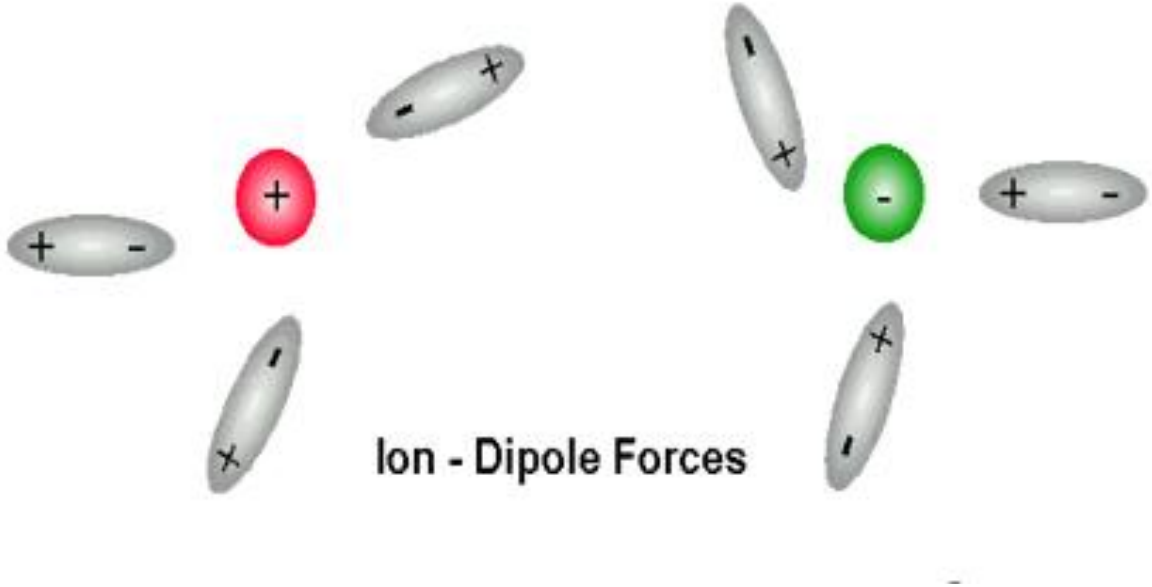
Intermolecular are attractive forces between molecules

- **Ion- Dipole forces**
- **Dipole-Dipole forces**
- **London dispersion forces**
- **Hydrogen bonds**

➢ Ion- Dipole forces

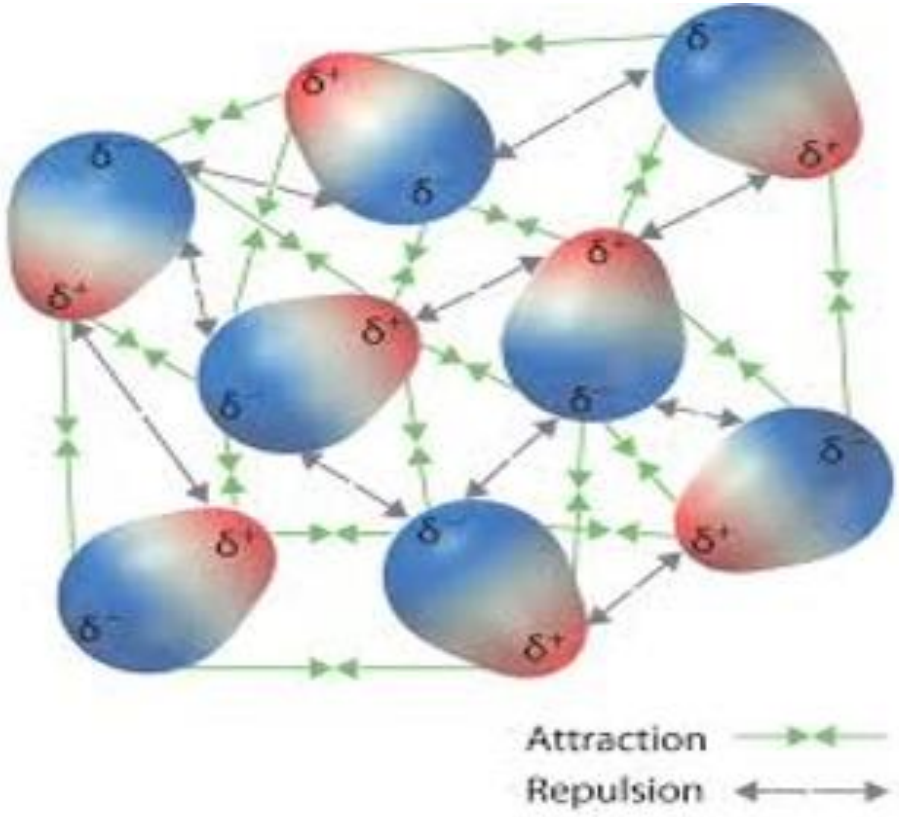
Occurs between ions and polar solvent

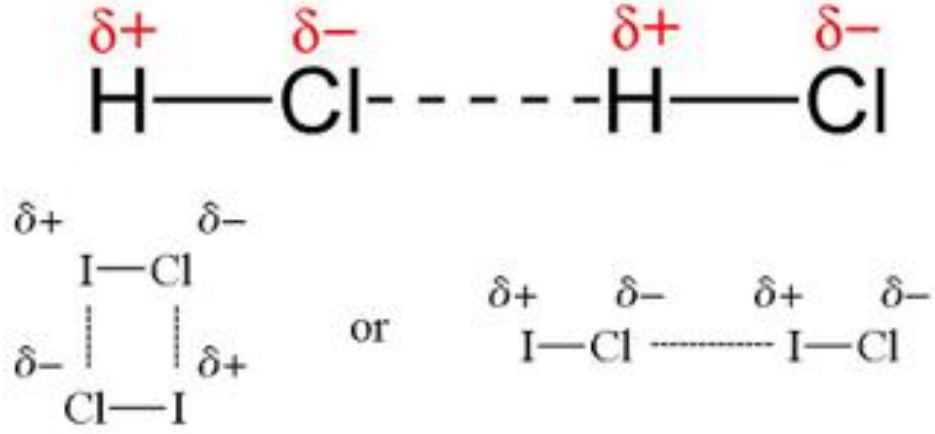




➢ Dipole-Dipole forces

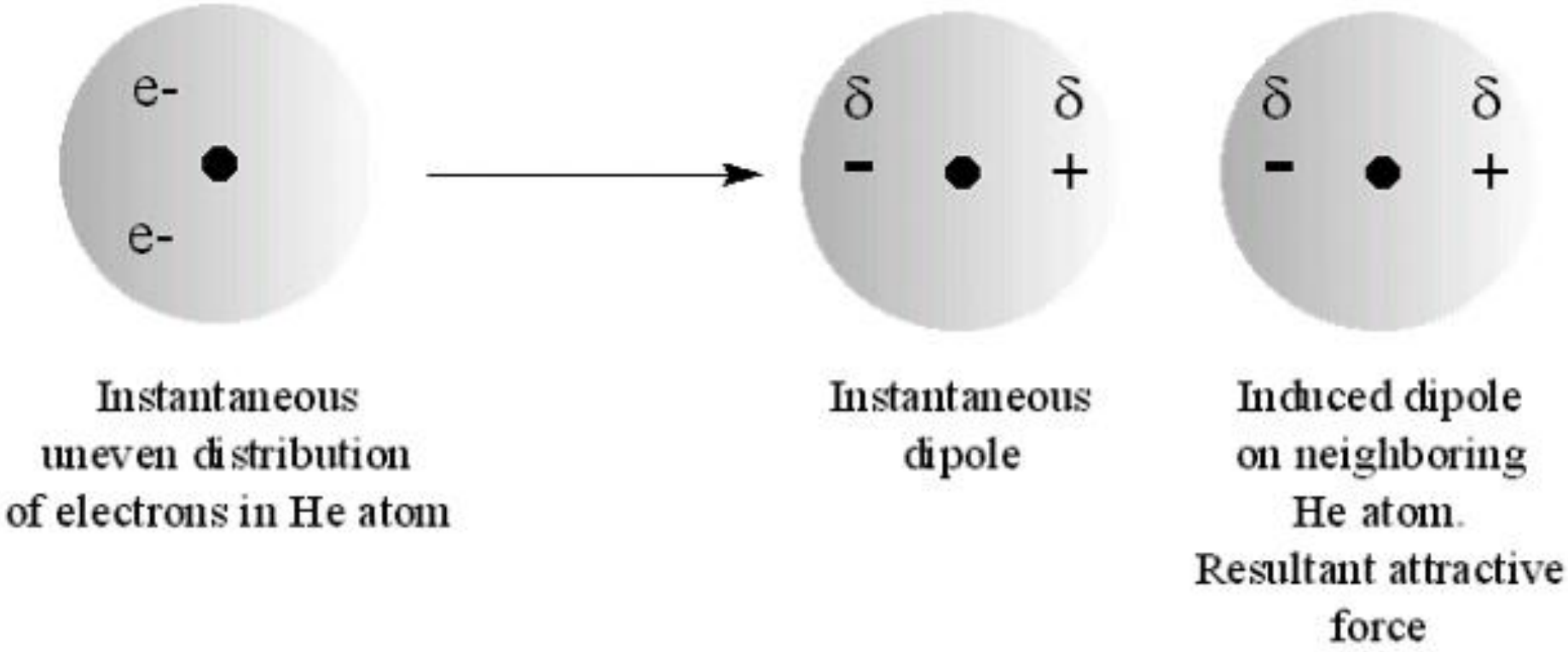
Occurs between polar molecules





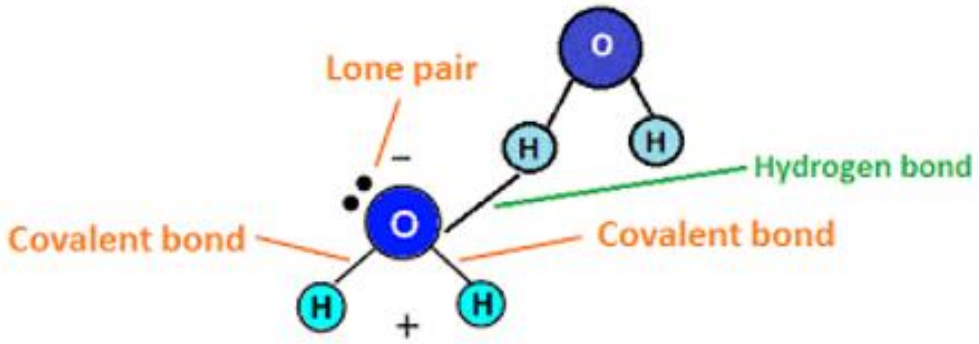
- **London dispersion forces**

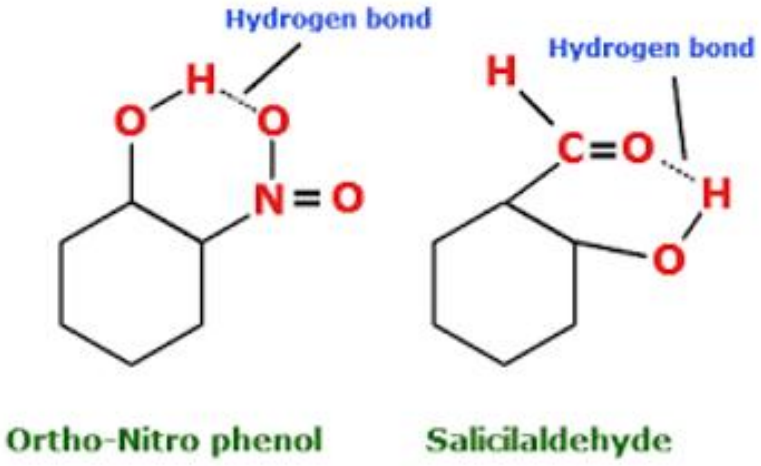
Occurs between all molecules; strength depends on size, polarizability

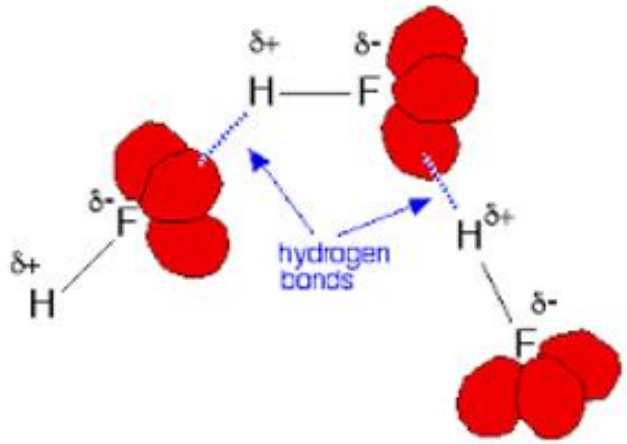


➢ Hydrogen bonds

Occurs between molecules with O-H, N-H, and F-H bonds

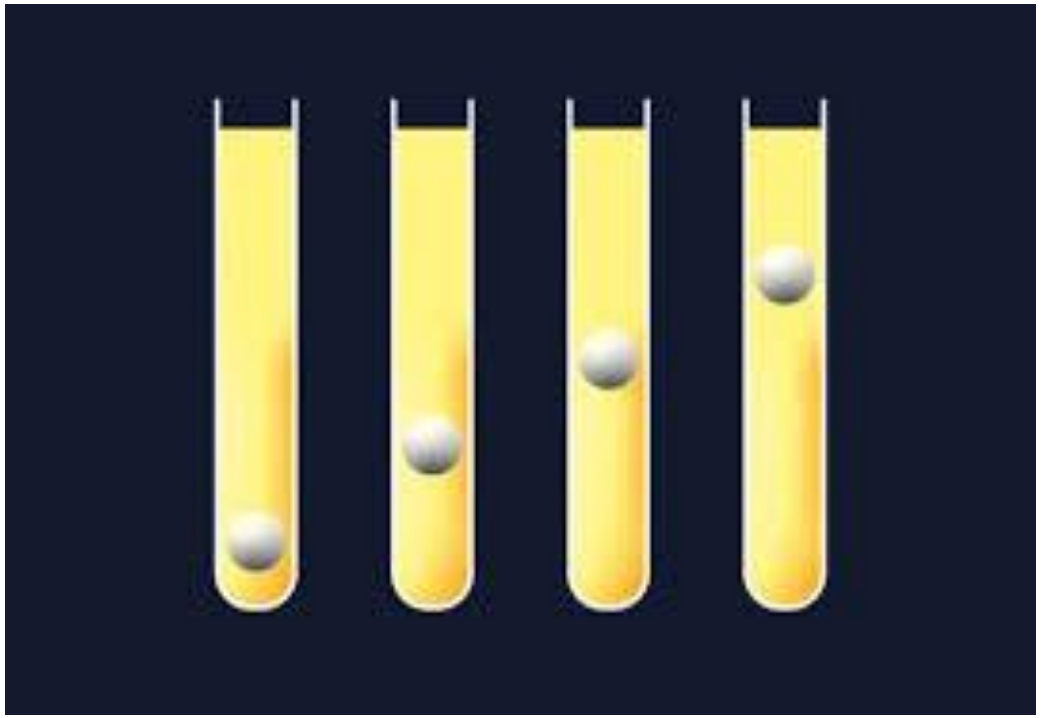


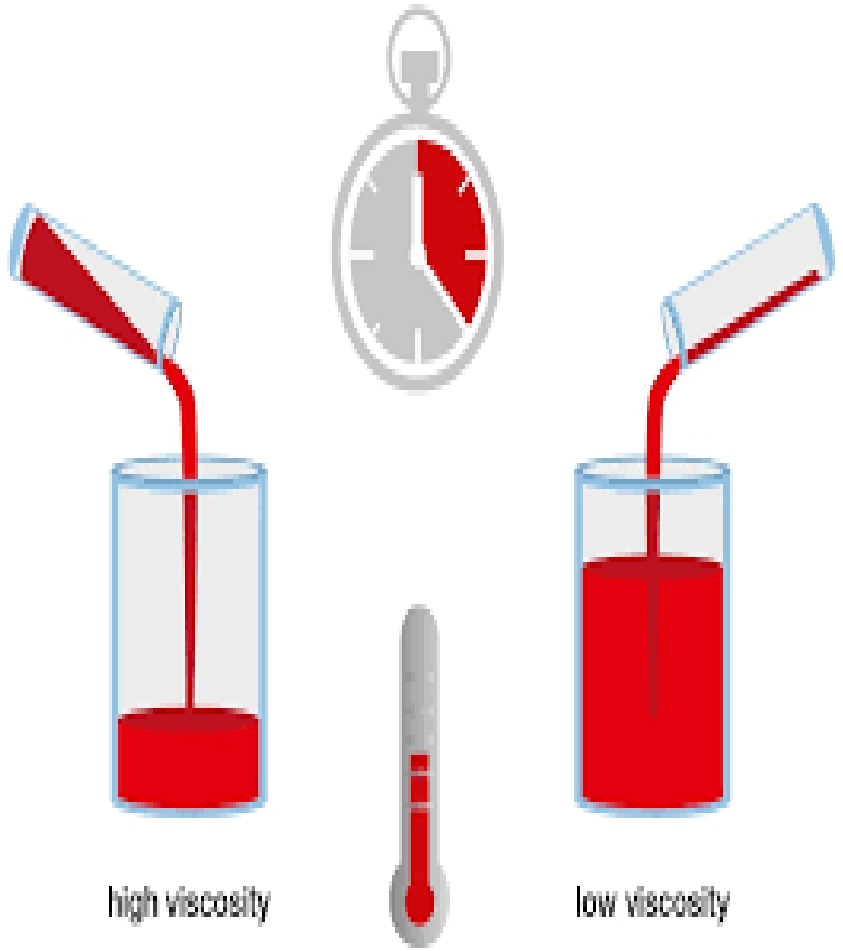




2.3. Some properties of liquids

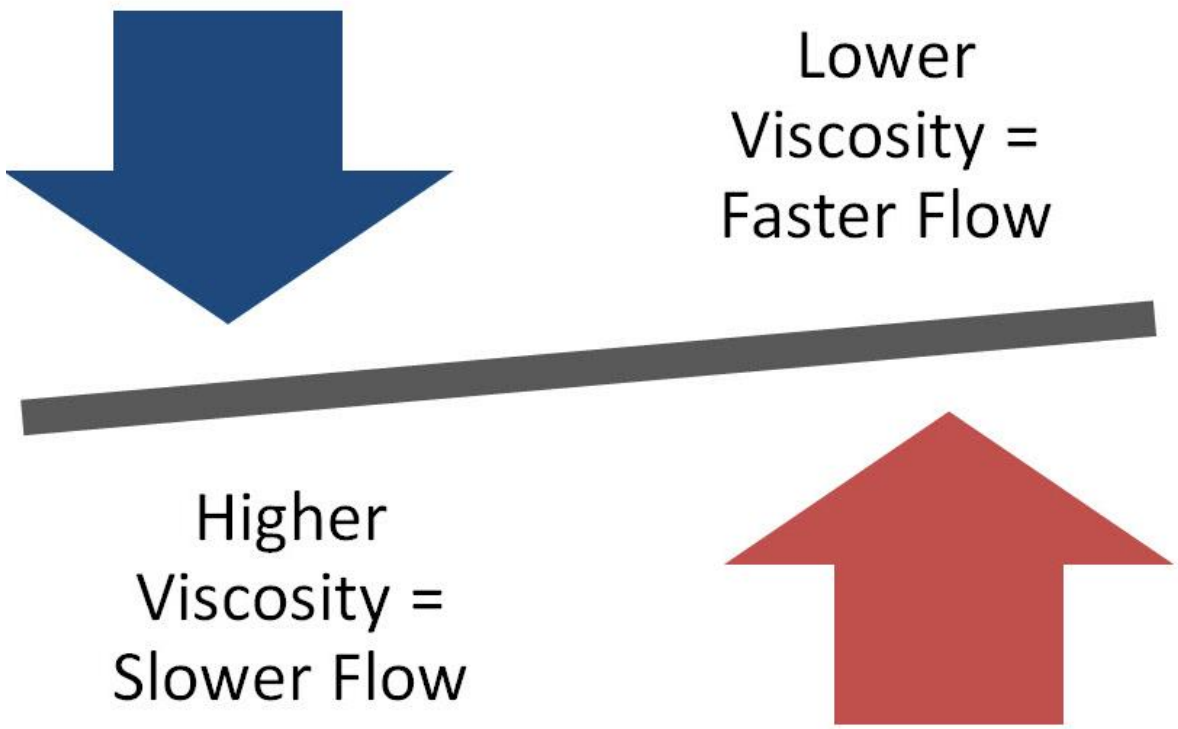
Viscosity: is the measure of a liquid's resistance to flow



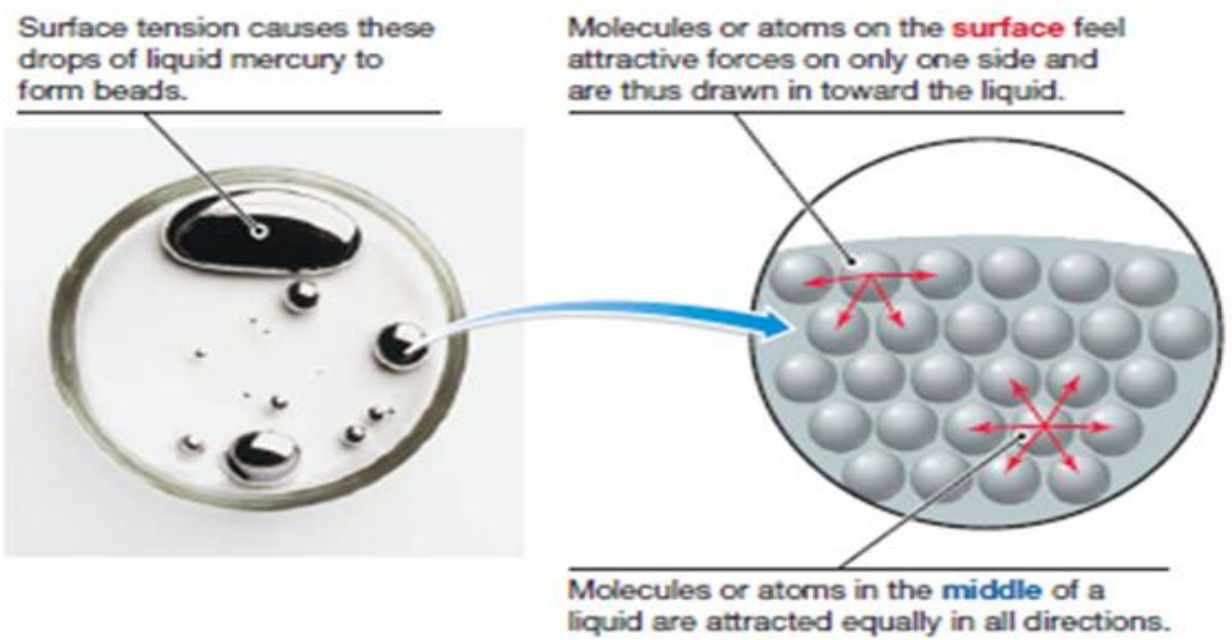


- Substances of nonpolar molecules, such as pentane and benzene have relatively low viscosities

- Polar substances, such as glycerol [$C_3H_5(OH)_3$] have higher viscosities



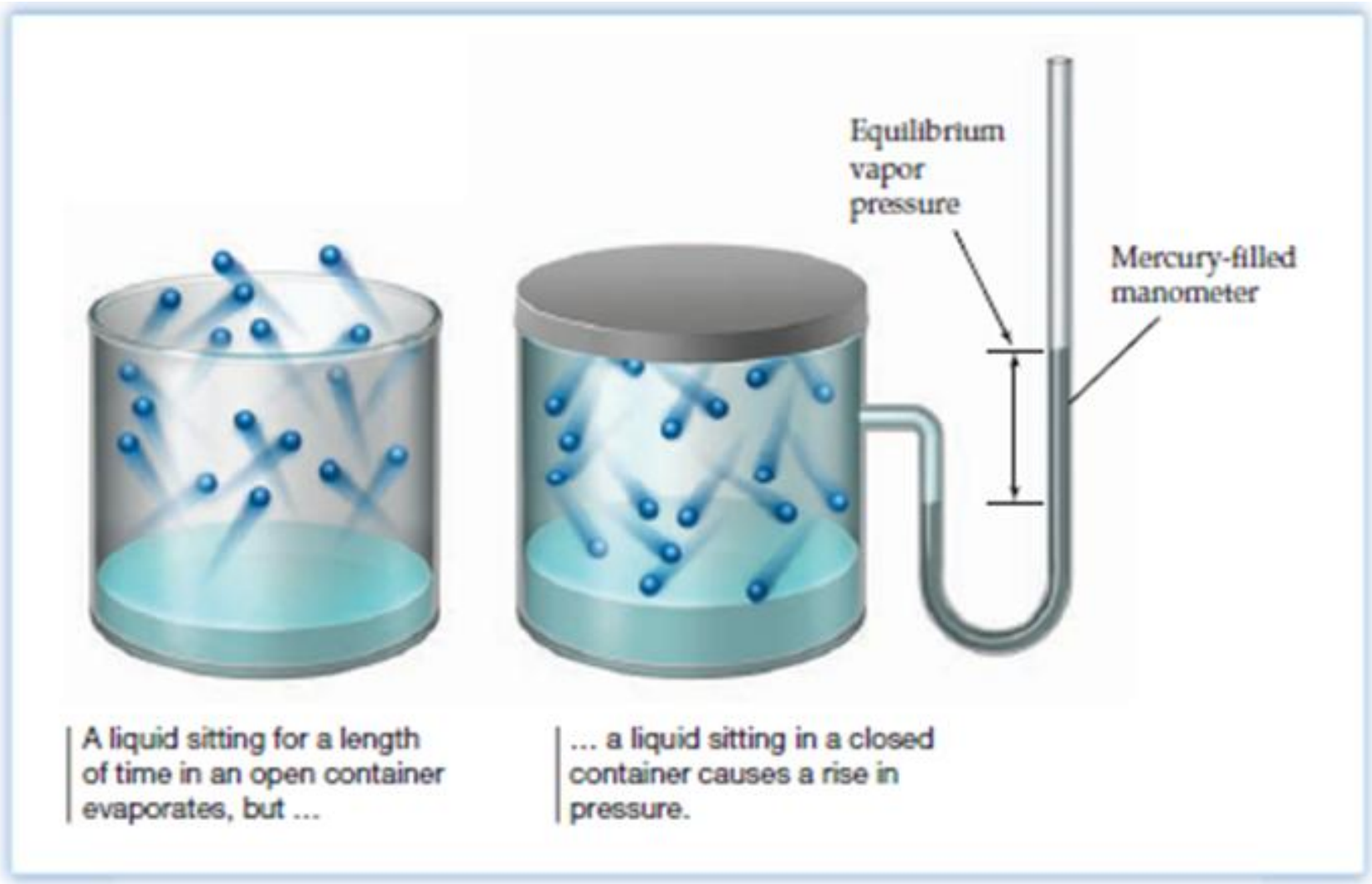
Surface tension: the resistance of a liquid to spread out and increase its surface area.



Notes

- Surface tension, like viscosity, is generally higher in liquids with stronger intermolecular forces.
- Both properties are also temperature-dependent because molecules at higher temperatures have more kinetic energy to counteract the attractive forces holding them together.

2.4. Evaporation, vapor pressure and boiling point



- Evaporation is when a liquid transitions to its gaseous state at temperatures below its boiling point.

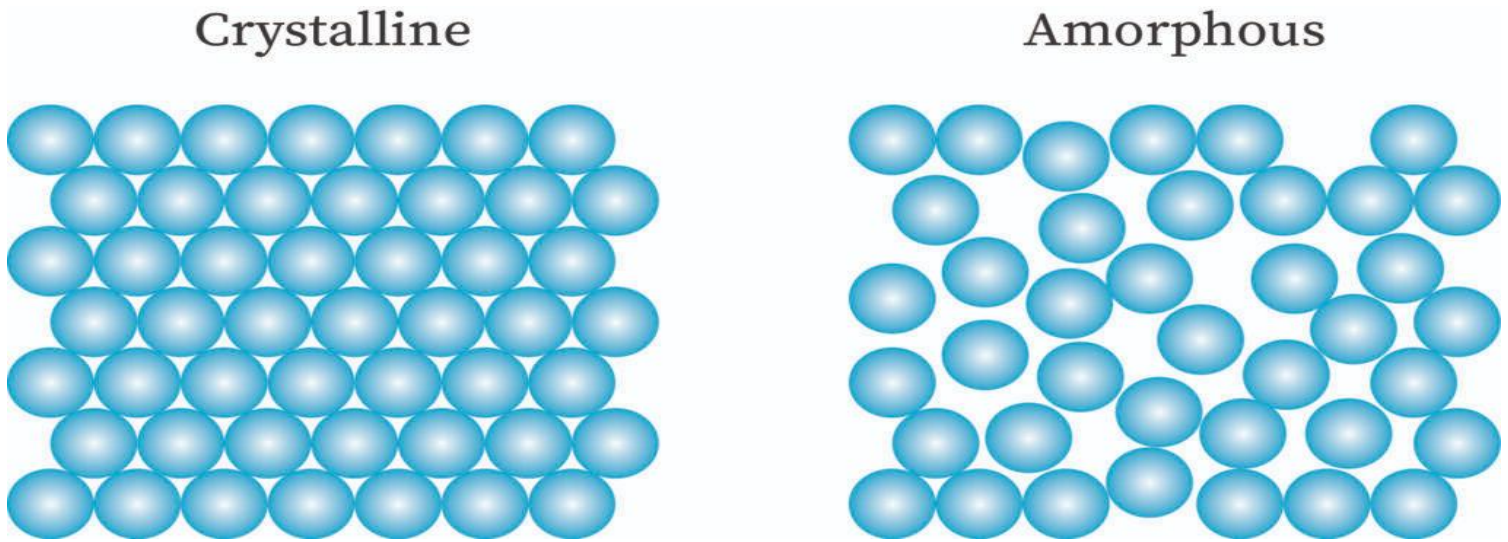
- Vapor pressure of liquid (Pvap), vapor pressure is the partial pressure of a gas in equilibrium.

- Boiling point of a pure substance is the temperature at which it transitions from a liquid to the gaseous phase (the temperature at which its vapor pressure equals the external pressure).

The boiling point at 1 atm is called the normal boiling point.

2.5. Kinds of solids

- Crystalline solids are those whose constituent particles atoms, ions, or molecules have an ordered arrangement extending over a long range.

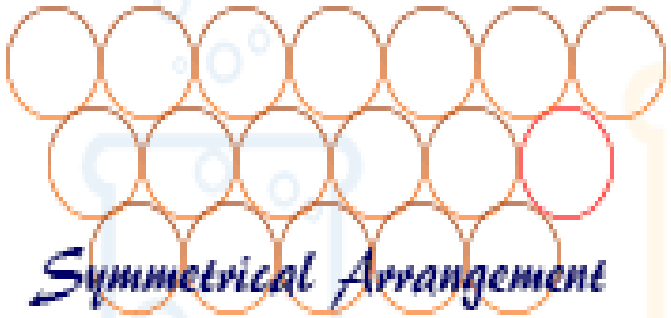


- Amorphous solids are those whose constituent particles are randomly arranged and have no ordered long-range Rubber is an example.

Comparison	Amorphous Solids	Crystalline Solids
Repeating units	Irregular	Regular
Geometrical shapes	No shapes	Exist
Melting points	Not definite	Fixed
Chemical nature	Isotropic	Anisotropic
Intermolecular forces	Not equally shared	Equally shared
Physical nature	Pseudo solids	True solids
Order of symmetry	Unsymmetrical	Symmetrical
Rigidity	Less	More
Range of ordered molecules	Short ordered range	Long ordered range
Examples	Glass, waxes, and plastic, etc	Diamond, metals, table salts, etc

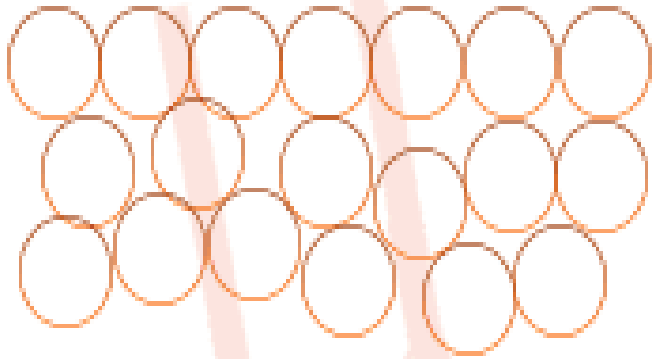
Types of Solids

Crystalline Solids



Symmetrical Arrangement

Amorphous Solids

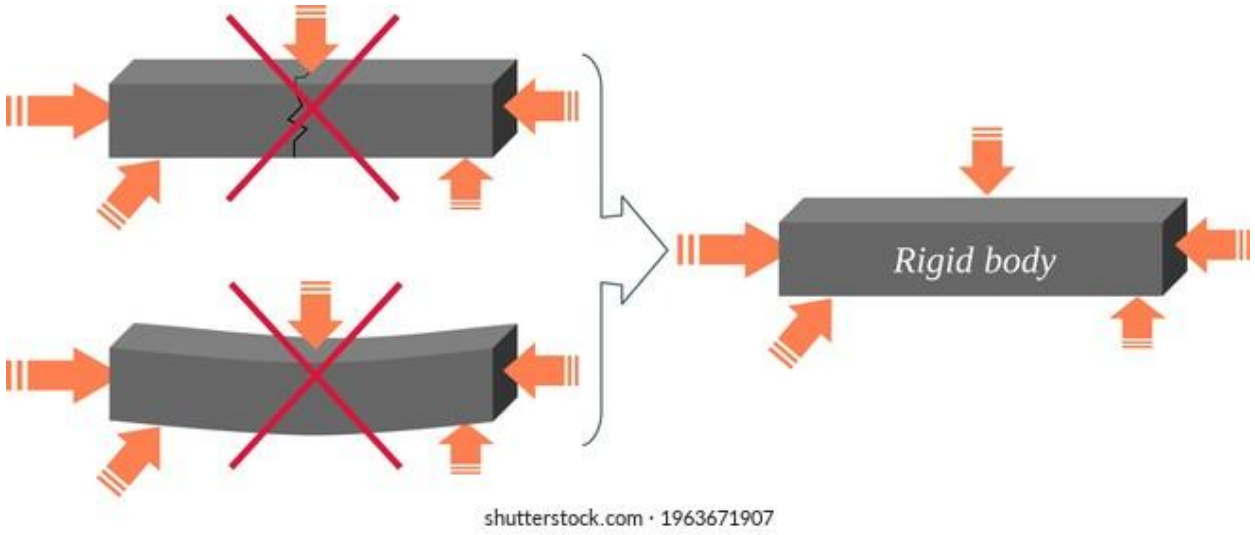


- Molecular Crystals
- Ionic Crystals
- Covalent Crystals
- Metallic Crystals

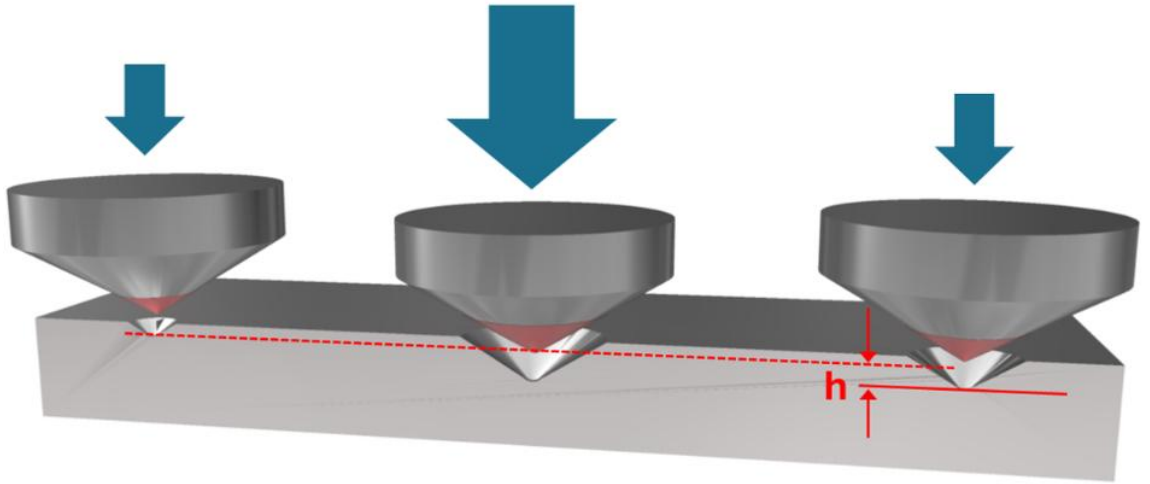
- **Molecular solids** are those like sucrose or ice, whose constituent particles are molecules held together by intermolecular forces
- **Ionic solids** are those like sodium chloride (NaCl), whose constituent particles are ions.
- **Covalent solids** are those like quartz or diamond, whose atoms are linked together by covalent bonds.
- **Metallic solids** such as silver or iron, also consist of large arrays of atoms, but their crystals have metallic properties such as electrical conductivity.

2.6. Solid Properties

- **Rigidity** in solids is the physical property that allows a solid object to resist changing its shape or size when an external force is applied.



- **Hardness** is a material's resistance to permanent surface damage, scratching, or localized plastic deformation



- **Melting point** is the exact temperature at which a solid substance changes into a liquid

